

Customer Shopping Behavior Analysis

Uncovering spending patterns, customer segments, and product preferences across **3,900 transactions** to guide strategic business decisions.



Dataset at a Glance

3,900

Total Purchases

Transactional records analyzed

18

Features

Columns covering demographics, purchases & behavior

37

Missing Values

In Review Rating — imputed by category median

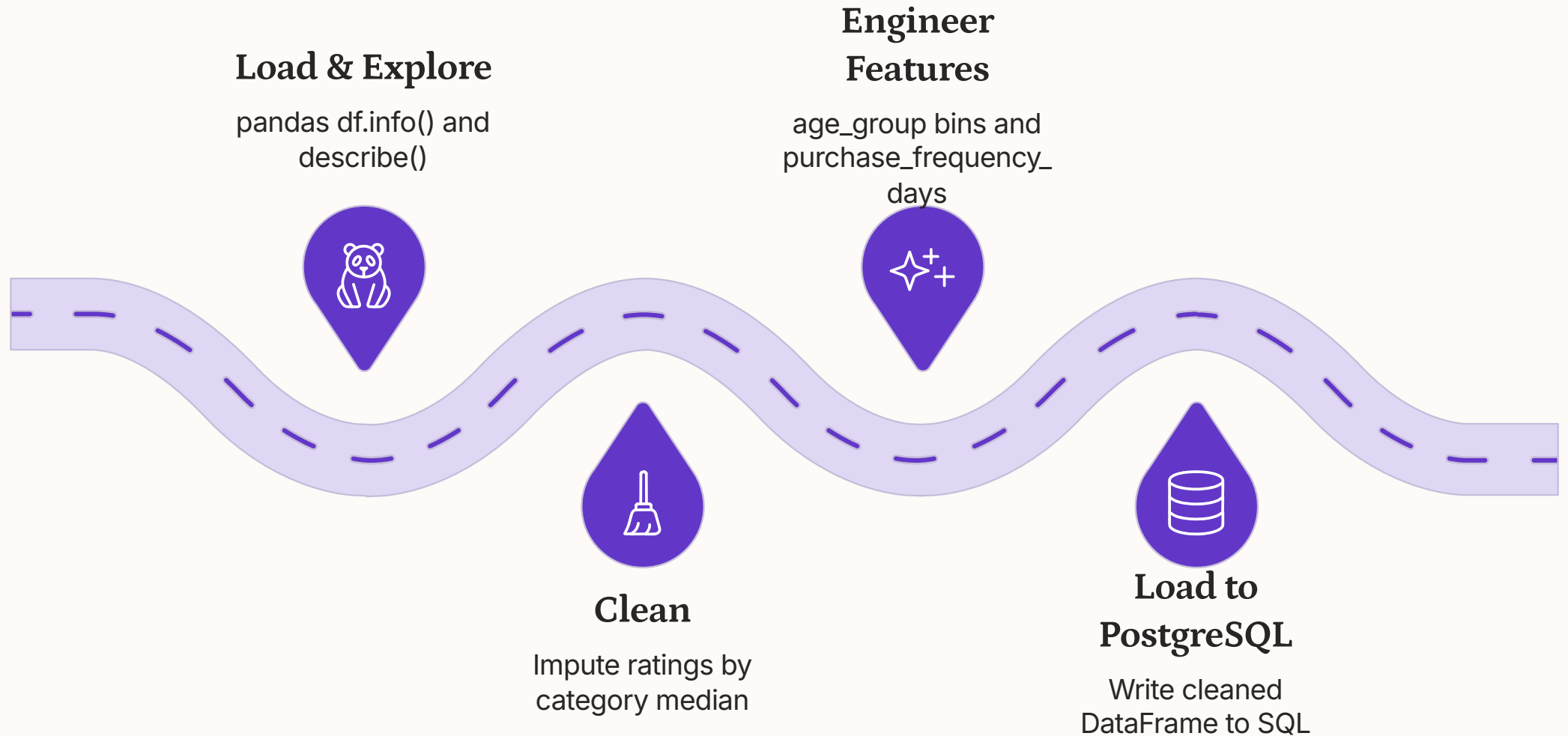
Key Feature Groups

- **Demographics:** Age, Gender, Location, Subscription Status
- **Purchases:** Item, Category, Amount, Season, Size, Color
- **Behavior:** Discount, Promo Code, Frequency, Review Rating, Shipping Type

The screenshot displays a data table within a software interface. The table has columns labeled A through H and rows numbered 1 through 23. The data is organized into groups, with some rows highlighted in different colors (e.g., red, green, blue, yellow). The table includes various numerical and categorical values, some of which are missing (indicated by empty cells). The interface also shows a sidebar with icons for different data manipulation functions and a top menu bar with various options.

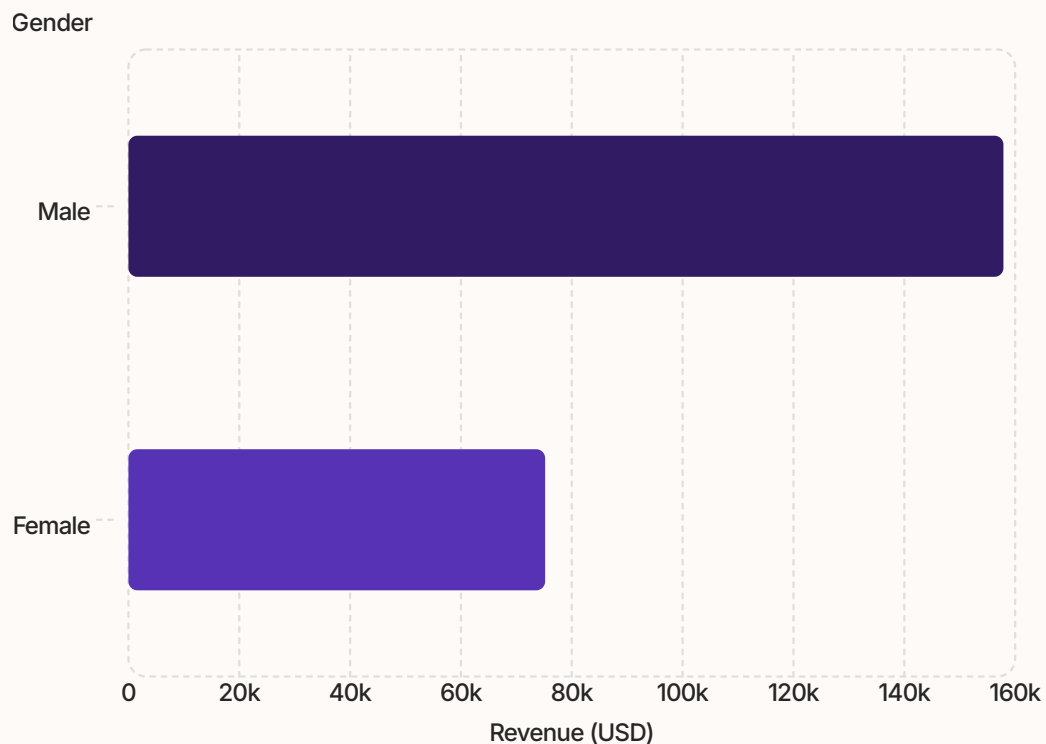
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4	Sphvia02.60	38	1.46	2.56		10		128								5
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10	E. I	120	4,22		10	11		00								

Data Preparation in Python



Columns were standardized to **snake_case**, and `promo_code_used` was dropped after confirming redundancy with `discount_applied`. The cleaned dataset was then loaded into PostgreSQL for structured analysis.

Revenue & Spending Insights



Gender Revenue Gap

Male customers generated **2.1× more revenue** than female customers (\$157,890 vs \$75,191).

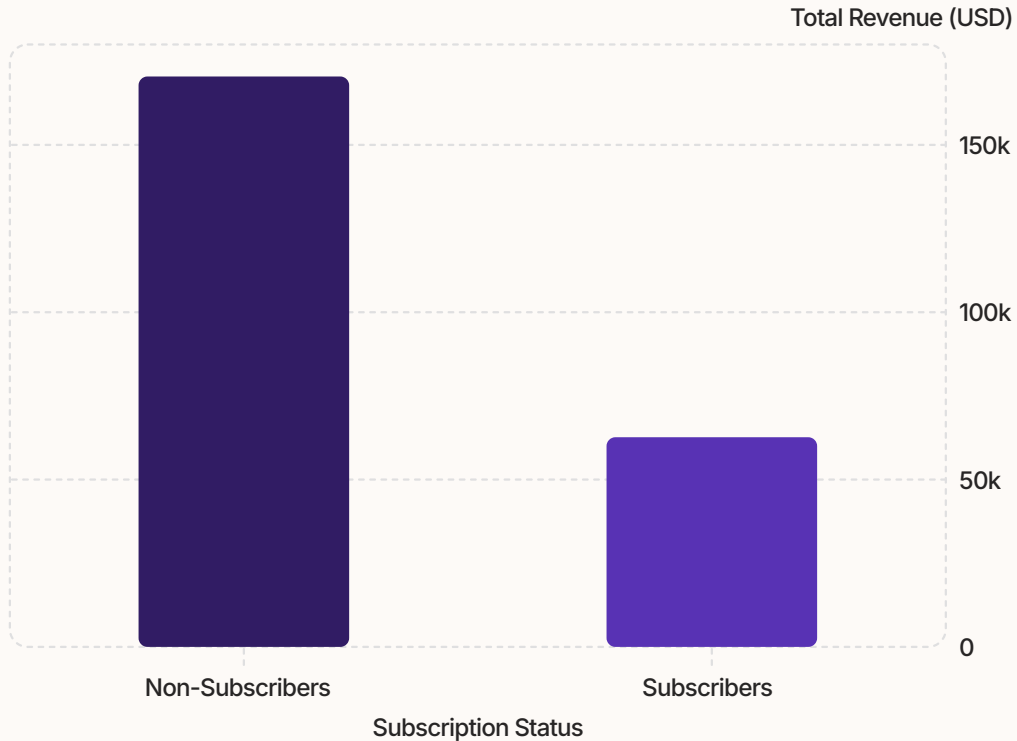
Discount-Dependent Products

Hat (50%), Sneakers (49.66%), Coat (49.07%), Sweater (48.17%), and Pants (47.37%) had the highest share of discounted purchases.

High-Spending Discount Users

839 customers used discounts yet still spent above the average purchase amount of **\$59.76**.

Subscribers vs. Non-Subscribers



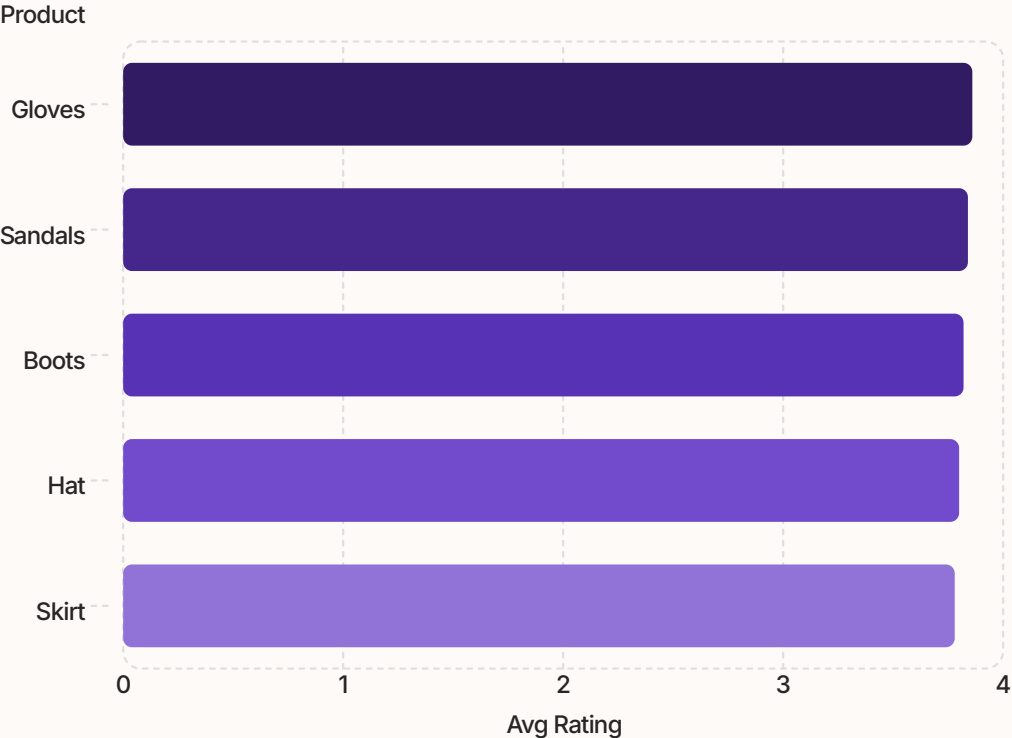
Key Findings

- Only **17.56%** of customers are subscribers (1,053 vs 2,847 non-subscribers)
- Average spend is nearly identical: \$59.49 (subscribers) vs \$59.87 (non-subscribers)
- Non-subscribers contribute **73%** of total revenue
- Of repeat buyers (5+ purchases), 958 subscribe vs 2,518 who don't

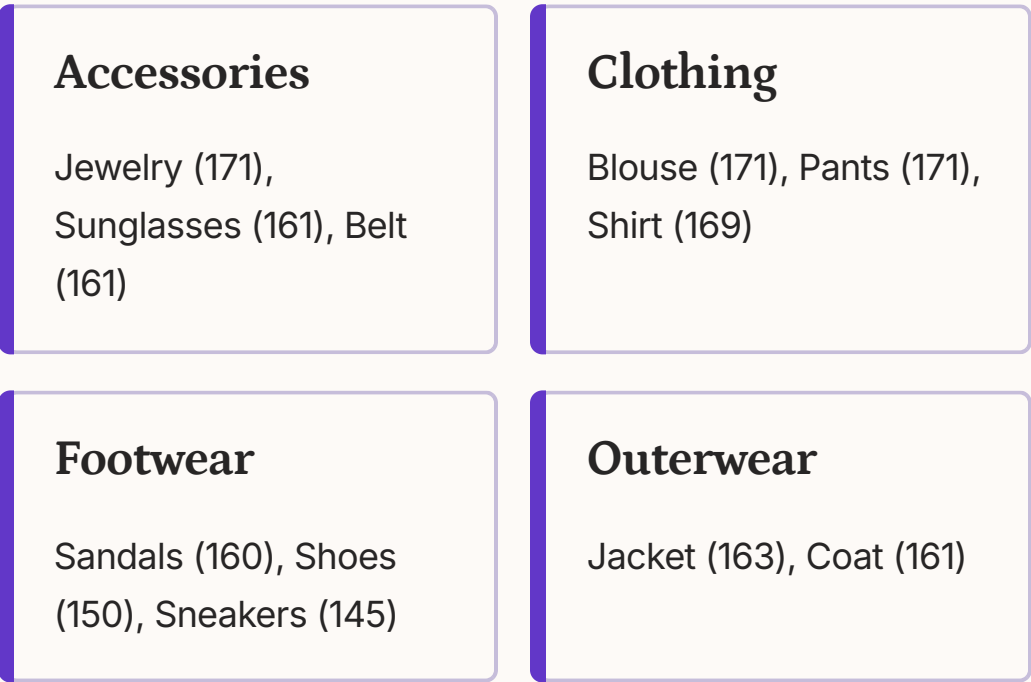
i Subscribers spend similarly but are far fewer — a major growth opportunity.

Product Performance

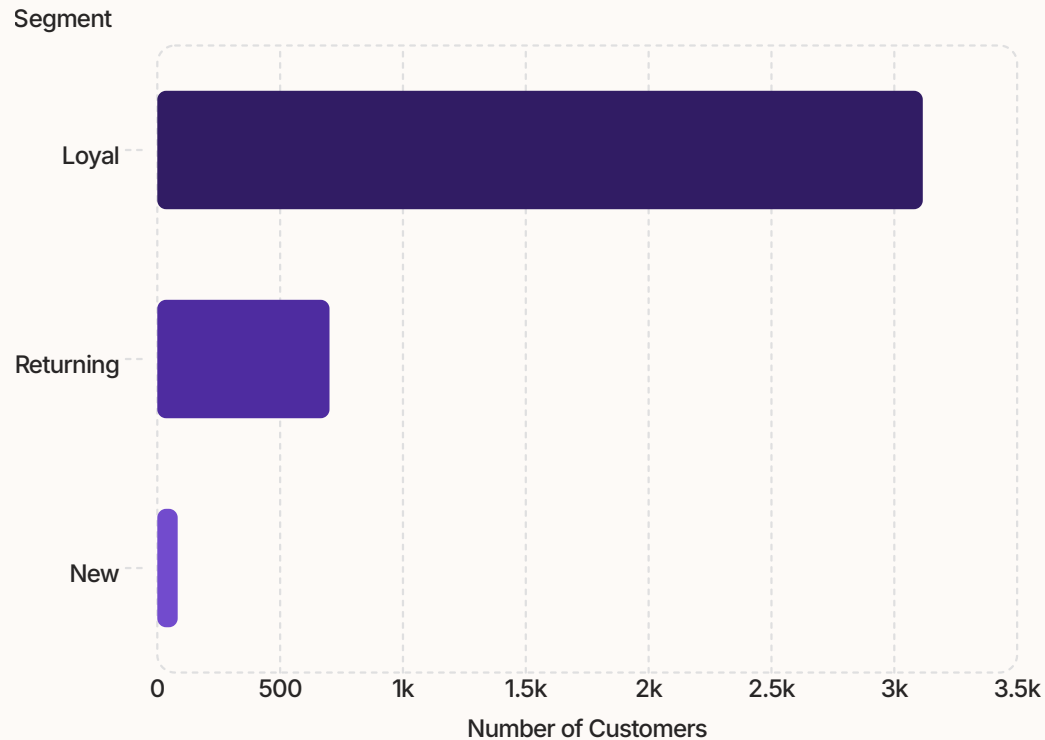
Top 5 Products by Review Rating



Top Products by Category



Customer Segmentation



Segment Breakdown

Loyal — 80%

3,116 customers with strong repeat purchase history

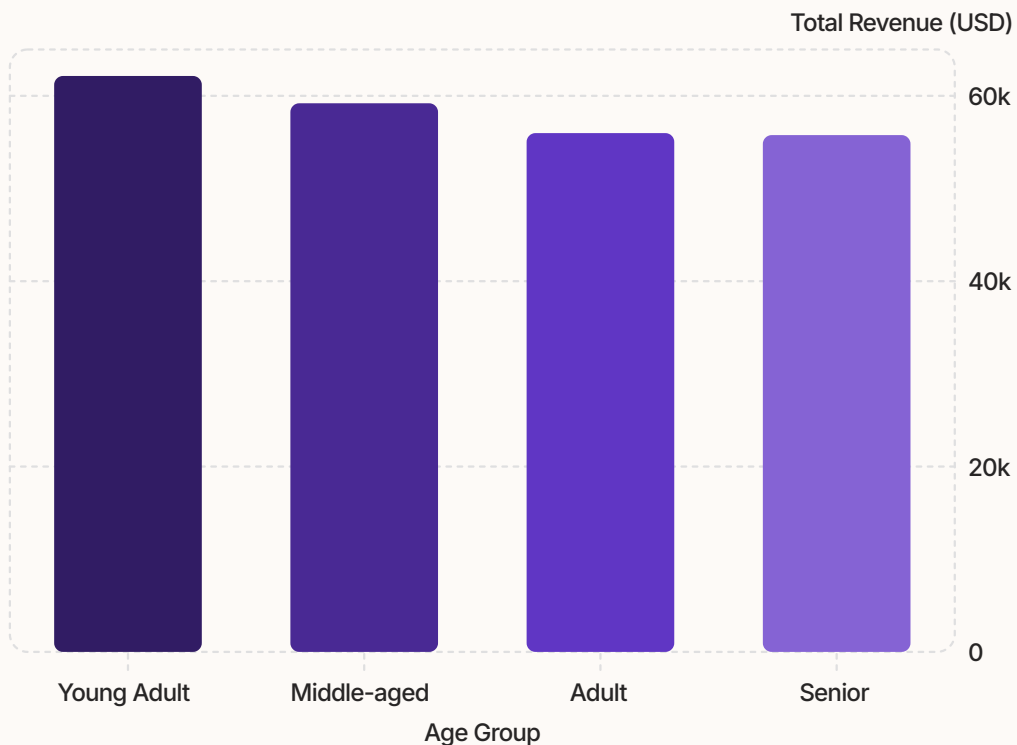
Returning — 18%

701 customers showing growing engagement

New — 2%

Only 83 first-time buyers — acquisition is a gap

Revenue by Age Group



Age Group Insights

Revenue is relatively balanced across age groups, with **Young Adults** leading at \$62,143 and **Seniors** close behind at \$55,763.

Shipping Behavior

Express shipping customers spend slightly more on average (**\$60.48**) vs Standard shipping (**\$58.46**), suggesting higher purchase intent.

Power BI Dashboard

An interactive dashboard visualizes key metrics for stakeholders at a glance.

Avg Purchase Amount

\$60.46 across all transactions

Avg Review Rating

3.77 out of 5 stars

Customers Analyzed

452 unique customers profiled

Top Revenue Category

Clothing leads at ~\$10K revenue



Business Recommendations



Boost Subscriptions

Promote exclusive subscriber benefits — only 17.56% are subscribed despite similar spend.



Loyalty Programs

Reward repeat buyers to convert Returning customers into the Loyal segment (80% of base).



Review Discount Policy

Balance sales lift with margin control — 5 products have ~48–50% discount rates.



Targeted Marketing

Focus on high-revenue age groups (Young Adults, Seniors) and Express shipping users.

